

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Common Types of Corporate Financing

- **Debt**, lending to the company
 - Main forms, bonds and loans
 - Cash flow, a fixed promised amount
 - “Senior”, paid back before equity in bankruptcy
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 - “Senior”, paid back before equity in bankruptcy
 - No voting rights or control (unless in bankruptcy)
- **Equity**, stock ownership in the company
 - Cash flow, dividends (and buybacks)
 - “Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout
 - Voting rights

Common Types of Corporate Financing

	Debt	Equity
Payment	Fixed promised payment	Residual (whatever is left over)
Priority in bankruptcy	Highest	Lowest
Risk	Low	High
Expected return	Low	High
Voting and control rights	No	Yes
Term	Dated	Perpetual
Share of total U.S. corporate financing (market values)	About 25%	About 75%

Module 5, Part 1

Bonds

Bonds, Definitions

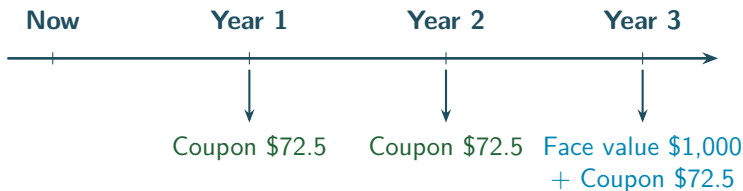
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Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$72.5 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value (par value, principal)**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the value (price) of the bond

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- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$

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- **Floating-rate bonds**, bonds whose coupon payments fluctuate with short-term interest rates (the coupon is not fixed)

Bonds vs. Loans, Main Differences

Bonds

- Sold to a broad group of investors, typically mutual funds, pension funds, insurance companies
- Subdivided into “smaller pieces”, usually \$1,000 face value
- Intended to be tradable
- Few “covenants” (restrictions) on the borrower

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Loans

- A single source of lending, typically a bank or a bank “syndicate” (group of banks)
- Not designed to be subdivided, although investors can still own “pieces” of a loan
- No active market, so harder to trade between investors
- Usually many covenants on the borrower

Bonds, Pricing

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- E.g., for a risk-free bond with time to maturity T ,

$$\text{Price} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

Bonds, Pricing a Risky Bond

- Risky bond, discount the **expected** coupon and face value payments, not the promised ones
 - The expectation accounts for the probability that the borrower does not pay

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- The required rate of return r is set by the opportunity cost, “what could I get investing in something else with a similar risk”
 - Depends on the time value of money and the riskiness of the bond (more on this a bit later)

How to Calculate Expectations?

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- The coupon is only *promised*, if the borrower defaults, it may pay only part of the coupon, or nothing
- **Expectation operator** $E[\cdot]$, the probability-weighted average payment the borrower will pay (which is also what the investor will receive)

Example, Expected Coupon

- Suppose in the first year, the borrower will pay,
 - \$72.50 (the full coupon) with 60% probability
 - \$50 with 20% probability (default but high “recovery”)
 - \$0 with 20% probability (default and no “recovery”)
- What is $E[\text{Coupon}_1]$?

Example, Expected Coupon

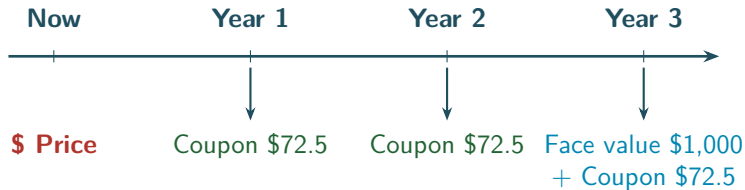
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- Solution, the expected coupon is the probability-weighted average amount,
$$E[\text{Coupon}_1] = 0.6 \times 72.50 + 0.2 \times 50 + 0.2 \times 0 = \$53.50$$

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- This is what goes in the numerator when discounting (risky) bonds

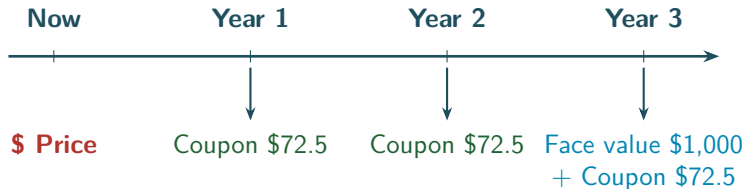
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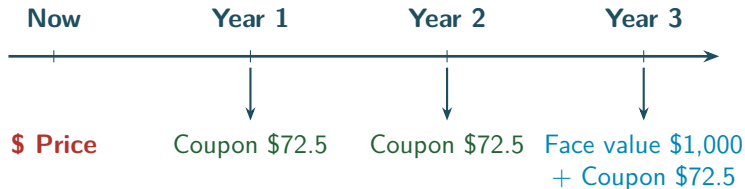
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$$\text{Price} = \frac{\$72.50}{(1 + 0.04)^1} + \frac{\$72.50}{(1 + 0.04)^2} + \frac{\$72.50 + \$1,000}{(1 + 0.04)^3} = \$1,090.19$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$72.5, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to price the bond

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- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

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- The output is $-\$1,090.19$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the price of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate, when **PMT** and **FV** are expected cash flows it is the required rate of return

Bond Prices Are Often Quoted as a Percentage of Par

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- E.g., the bond we just analyzed, face value \$1,000 and price \$1,090.19, could be quoted as,

$$\text{Price} = 109.019\%$$

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors

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- In return, the companies make pre-specified payments of coupons and face value to the investors

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The price (value) of a bond is the discounted (expected) payments, at the required rate of return

Bond Yields

- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's promised payments equals its price (also simply the “yield”)

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- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's promised payments equals its price (also simply the “yield”)
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield to maturity by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 7.25%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,090.19. What is its YTM?

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 - **CPT**, then **I/Y**
- The yield is 4%

Bond Prices Are Often Quoted as Yields

- For each bond, price and yield have a direct one-for-one relationship

U.S. 10 Year Treasury

US10Y:Tradeweb

RT Quote | Exchange

Yield | 5:05 PM EST

1.822% ▼ **-0.162**

1D 5D 1M 3M 6M YTD 1Y 5Y ALL



SUMMARY

NEWS

KEY STATS

Yield Open	1.892%	Price	100.4844	Price Day High	100.5312
Yield Day High	1.927%	Price Change	+1.4688	Price Day Low	99.5312
Yield Day Low	1.816%	Price Change %	+1.4844%	Coupon	1.875%
Yield Prev Close	1.984%	Price Prev Close	99.0156	Maturity	2032-02-15

LATEST ON U.S. 10 YEAR TREASURY

- Oil prices could determine how markets react to Russia's Ukraine invasion 49 MIN AGO - CNBC.COM
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- So bond prices are often quoted not in dollars or percent of par, but as the “yield”

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RT Quote | Exchange

Yield | 5:05 PM EST

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1D 5D 1M 3M 6M YTD 1Y 5Y ALL

+ Comparison

1D

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- So bond prices are often quoted not in dollars or percent of par, but as the “yield”
- E.g., the 10-year U.S. Treasury bond is quoted as a yield of 1.822% (its price is 100.48%)

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RT Quote | Exchange

Yield | 5:05 PM EST

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Is the YTM the Same as the Required Rate of Return r ?

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- For risk-free bonds, YTM and r are the same
- For risky bonds, YTM and r are different (specifically, $\text{YTM} > r$)

YTM vs. r , Promised vs. Expected Payments

- When calculating r , use **expected** payments, $E[\cdot]$ accounts for the probability that the borrower pays less or nothing,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \frac{E[\text{Coupon}]}{(1+r)^2} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

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- When calculating the YTM, always use **promised** payments, even for risky bonds,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?

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- B. The required return, using the **expected** cash flow,
 $(1 - p_{\text{default}}) \times \text{FV} = 900$

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- r is 5.88%

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- A bond priced **equal** to its face value trades at **par**
 - When will this happen?

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Example, Bond Premiums and Discounts

- A (risk-free) bond pays annual coupons. Its face value is \$1,000. Its time to maturity is five years. Its yield is 5%
- What is the price of the bond if the annual coupon rate is,
 1. 7%?
 2. 3%?
 3. 5%?

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 - 1000 **FV**, 5 **N**, 5 **I/Y**, 70 **PMT**, then **CPT** **PV**
- The price is \$1,087, a “premium bond”
- Bonds 2 and 3 are worth \$913 (a “discount bond”) and \$1,000 (“par”), respectively

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield to maturity
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield to maturity of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

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- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Practice Problem

- A firm's bonds pay an annual coupon rate of 6.8%. Their face value is \$1,000. Their price is \$795.69. Their yield to maturity is 9.50%
- What is the remaining maturity of these bonds?

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$

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- The remaining maturity is about 14 years ($N = 13.98$)

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Determinants of the Prevailing Interest Rates

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Determinants of the Prevailing Interest Rates

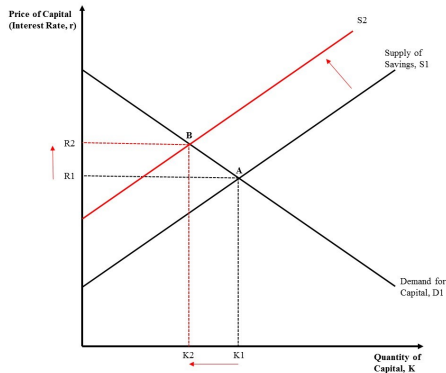
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- The Federal Reserve does not “set” the interest rate, it tilts supply and demand by buying and selling bonds (creating “money” to buy them)

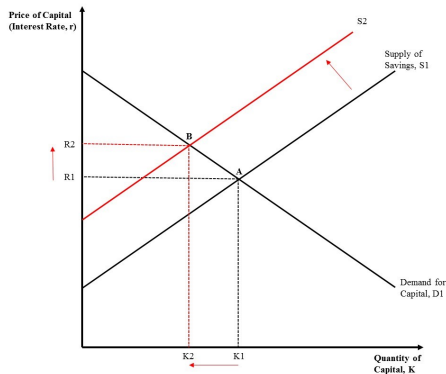
Example, Negative Supply Shock

- The market starts at equilibrium point A, interest rate R_1 and quantity of lending K_1



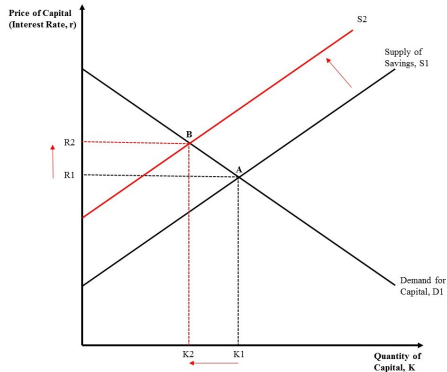
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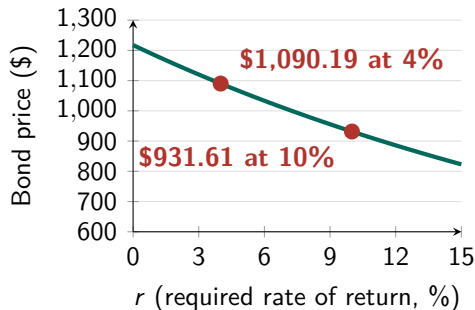
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- See also [Investopedia, Forces Behind Interest Rates](#)



Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

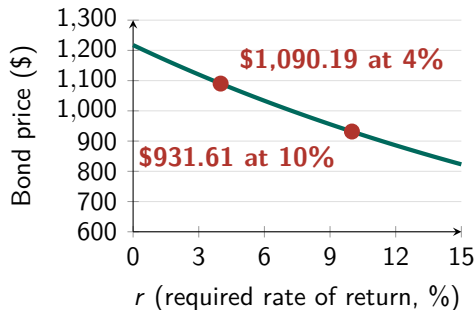


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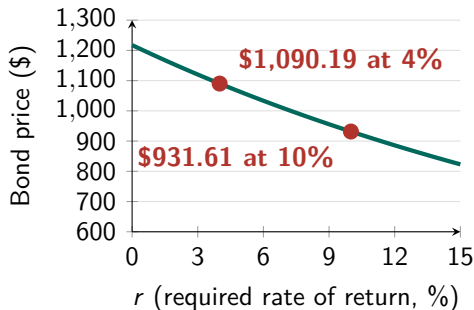


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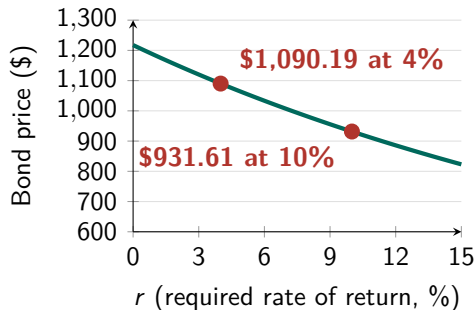


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- **Default premium**, the additional yield on a bond that investors require for bearing credit risk

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Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr	Safety
Investment-grade bonds			
Aaa	AAA	0.1%	The strongest rating, ability to repay interest and principal is very strong
Aa	AA	0.7%	Very strong likelihood that interest and principal will be repaid
A	A	2.2%	Strong ability to repay, but some vulnerability to changes in circumstances
Baa	BBB	3.5%	Adequate capacity to repay, more vulnerability to changes in economic circumstances

Source, Brealey, Myers, and Marcus, Table 6.2

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr	Safety
High-yield bonds			
Ba	BB	15.7%	Considerable uncertainty about ability to repay
B	B	35.5%	Likelihood of interest and principal payments over sustained periods is questionable
Caa, Ca	CCC, CC	48.9%	Bonds that may already be in default or in danger of imminent default
C	C	n/a	Little prospect for interest or principal on the debt ever to be repaid

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Default Risk and Bond Yields

Issuer	Ticker	Coupon (%)	Maturity	Rating	Yield (%)
Microsoft	MSFT	2.40	2022	AAA	2.89
Walmart	WMT	6.75	2023	AA	2.92
Bristol Myers	BMJ	6.80	2023	A	3.43
Corning	GLW	7.73	2024	BBB	4.00
Southwestern Energy	SWN	4.10	2022	BB	4.80
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- Yields tend to be higher (prices lower) for lower-rated bonds

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- It is the “holding period return” of the bond, as in [Module 4](#)

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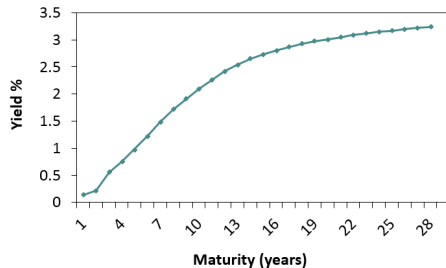
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- Then,

$$\text{Rate of return} = \frac{\text{Coupon income} + \text{Price change}}{\text{Investment}} = \frac{80 + (832.70 - 908.71)}{908.71} = 0.44\%$$

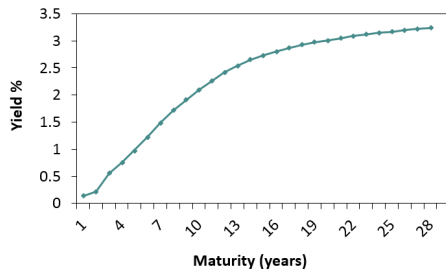
The Yield Curve

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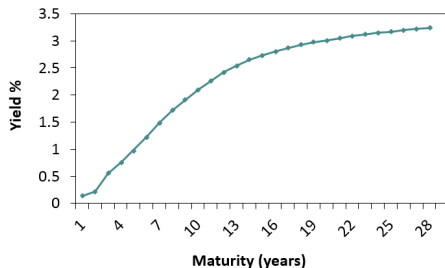
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- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 2

Stocks

Goals

- **Part 1, bonds**
 - Differences between debt and equity
 - Valuation
 - Yields
 - Returns and interest rates
 - Default risk
 - Special features of bond securities
- **Part 2, stocks**
 - Valuation, dividend discount model
 - Valuation, comparables
 - Market efficiency

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- The latter formula is often called the “**Gordon growth model**”

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- That is, dividend yield plus expected capital appreciation (in %)

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- Uncertain dividend forecasts
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- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies
 - But future dividends are then probably even more uncertain!

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

Practice Problem

- Consider the following three stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
 - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$P_C = \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} +$$

$$\frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6}$$

$$P_C = \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6$$

$$P_C = \$104.51$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity, $P_A = \$142.86$, $P_B = \$166.67$, and $P_C = \$156.50$, so Stock B is the most valuable

Valuing Stocks, Comparables

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

- Furniture manufacturer Herman Miller has earnings per share of \$1.38
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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

Example, Valuation by Comparables

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- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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- Assumes the comparables are priced correctly
 - Cannot tell if an entire industry is overvalued or undervalued
 - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

Buying and Selling Stock on the Stock Market

- Current stock price

FedEx Corporation (FDX)

NYSE - Nasdaq Real Time Price. Currency in USD

☆ [Add to watchlist](#)

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary

[Chart](#)

[Conversations](#)

[Statistics](#)

[Profile](#)

[Final](#)

Previous Close	249.02	Market Cap	64.318B
Open	246.00	Beta	1.57
Bid	239.98 x 100	PE Ratio (TTM)	22.11
Ask	240.10 x 100	EPS (TTM)	10.86
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)
Volume	2,359,358	Ex-Dividend Date	N/A
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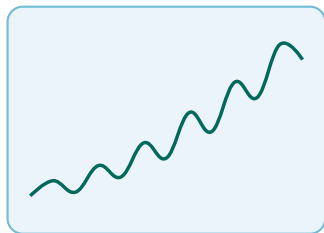
- Market capitalization
- **Beta**, a measure of risk (how much the stock moves with the overall market)
- Price/earnings
- EPS
- Dividend and dividend yield

Can You Beat the Stock Market?

- It's really hard!

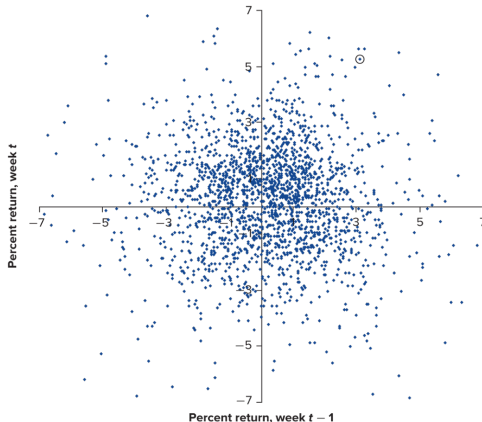
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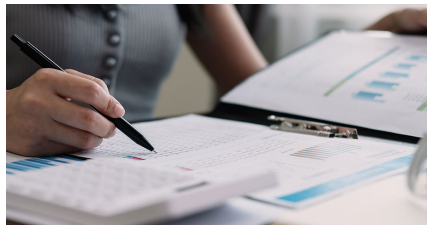
Discuss, Where Is the Pattern?

- Scatter plot of the NYSE Composite Index return in one week against its return in the previous week



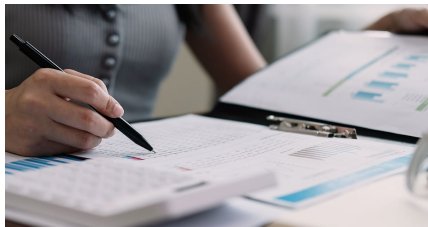
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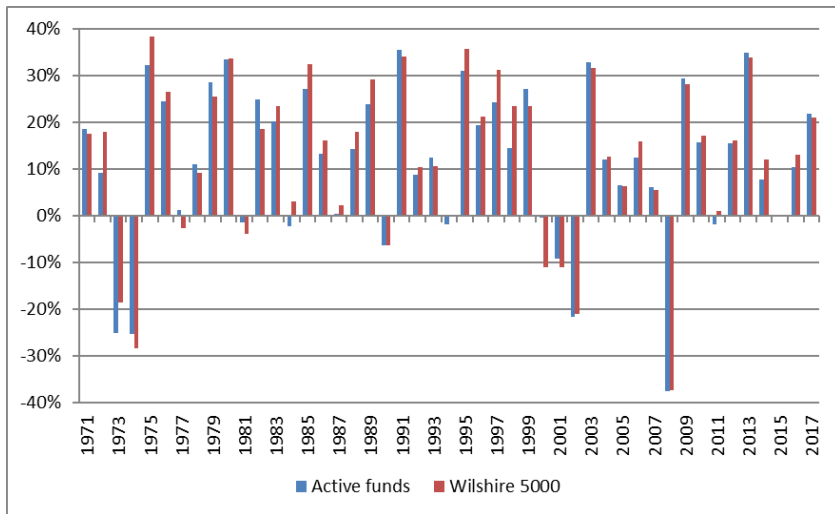


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Professional Fund Managers Don't Generally Beat the Index!



Information in Stock Prices

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- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”
- What **type** of information is captured in the stock price?
 1. All historical price information? (“weak” efficiency)
 2. All public, easily available information? (“semi-strong” efficiency)
 3. Also all private or difficult-to-interpret information? (“strong” efficiency)

What Does This Mean for Investors?

Pros 👍

- If stocks are fairly priced, even an investor who **knows nothing** can buy **any** stock and expect it to fairly compensate them for the risk they take

Cons 👎

- To *consistently* make money above the *fair* (*risk-compensating*) return by betting on publicly traded shares, an investor needs a very distinct competitive advantage
- E.g., expertise or access to information that only a few people can figure out

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- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Beating the market is really hard, prices already reflect available information (the efficient markets hypothesis)

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- Interest rates and bond prices move in opposite directions
- Stocks, the dividend discount model or comparables, and efficient markets make the index hard to beat

Thank you!

Please let me know if you have any questions.